MAPPA: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping traces atom identities between molecular endpoints, supporting reaction analysis and pathway construction. Yet the same endpoints can admit multiple correspondences, and selecting a single answer can conceal alternatives with different chemical implications. We introduce MAPPA, a framework that efficiently discovers and retains alternative atom mappings. It uses shared molecular fragments and molecular symmetry to represent these alternatives compactly, allowing their bond changes to be examined and supporting subsequent chemistry-aware selection. MAPPA achieves the highest reference coverage among the evaluated methods on Golden. Standard benchmark runs recover 99.08% of Golden and 99.63% of EnzymeMap reference mappings, with matching and decoding together averaging 3.19 and 13.16 CPU seconds per reaction, respectively. Broader exploration raises Golden coverage to 99.57%. These results make broad exploration of mapping alternatives practical, providing a foundation for studying reaction ambiguity and incorporating chemical knowledge into mapping selection.

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1 Introduction

Atom-to-atom mapping (AAM) assigns a correspondence between atoms at the two molecular endpoints of a reaction. It describes the net evolution of atom identities through a chemical transformation, leaving the intervening elementary steps unspecified. Once this correspondence is established, changes in bonding can be identified explicitly (Figure 1). These assignments underpin reaction-center identification and the extraction of transformation templates used in reaction prediction and retrosynthesis (14; 1). They also connect endpoint geometries for reaction-path calculations (3). Reliable mapping therefore matters both for interpreting individual reactions and for building the datasets and models used to study chemical reactivity.

Endpoint structures, however, do not generally determine a unique correspondence. Molecular symmetry can make several atom labels structurally equivalent, while other assignments imply different atom origins and bond changes. Favoring fewer bond changes helps narrow the choices, but need not select the annotated mapping. In the EnzymeMap example in Figure 2, the reference places the formaldehyde carbon in the three-carbon molecule, whereas another correspondence assigns it to CO_2. Their heavy-atom counts differ, but both require six total bond events when hydrogen assignments are optimized separately for each correspondence. Both also account for the complete endpoint composition. Event counts therefore leave distinct atom origins unresolved in this example. Retaining alternatives allows subsequent interpretation to draw on additional chemical knowledge.

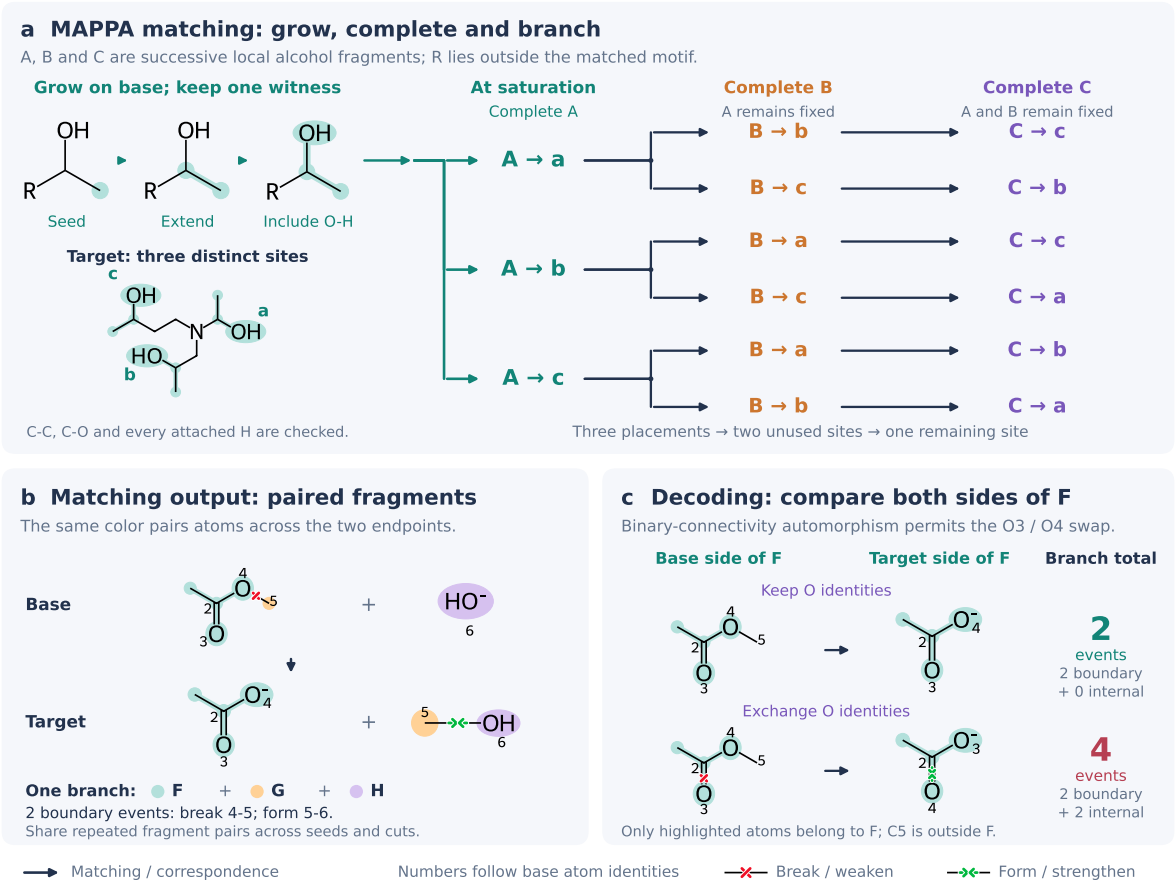


Figure 1 Fragment-based exploration of atom mappings with MAPPA. (a) Greedy growth and conditional placements. (b) Colors pair the base and target atom sets of fragments F, G and H. (c) An automorphism of F's binary connectivity permits O3 and O4 to exchange. Original bond orders then give two internal events, added to the two boundary events in b. Both endpoints are shown; unhighlighted C5 is outside F. Numbers follow base atom identities. Red marks indicate cleavage or weakening, green marks formation or strengthening. Constructed examples are detailed in Appendix A.

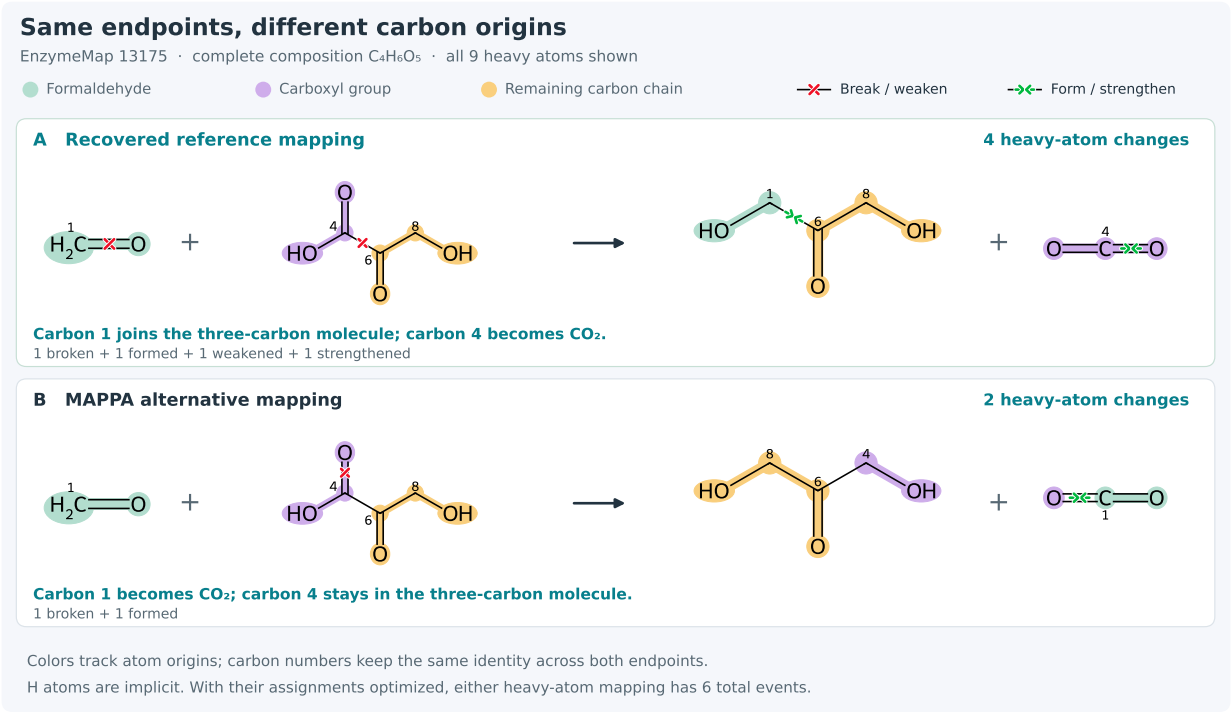


Figure 2 Distinct atom origins can share the same total event count. Two saved MAPPA heavy-atom correspondences for EnzymeMap case 13175 retain all nine heavy atoms and the complete composition $C_4H_6O_5$. A recovers the annotated mapping: formaldehyde carbon 1 enters the three-carbon molecule, while carboxyl carbon 4 becomes CO_2. B exchanges these carbon origins. The drawings show four and two heavy-atom bond changes, respectively; colors track atom origins and numbers identify the carbons. Minimizing hydrogen assignments separately for these fixed heavy-atom mappings adds two and four events, respectively, giving six total events in either case. Thus this example shows ambiguity under event counting, not a strict all-atom advantage over the reference. Hydrogen identities are not annotated by EnzymeMap. These correspondences do not establish reaction pathways. Provenance and counting details are given in Appendix A.1.

AAM methods can be organized by the output they seek to provide (Table 1). *Alternative-generating methods* retain several correspondences for subsequent interpretation or selection. SLAPMapper provides such alternatives through approximate graph matching, combining iterative refinement of atom labels with sequential linear assignment (9). *Prediction-oriented methods* aim to select a preferred correspondence. RXNMapper derives assignments from attention patterns in a reaction language model trained on unannotated reactions (14); LocalMapper learns from chemist-labeled examples through a human-in-the-loop procedure (1). These methods bring information from reaction data into the selection of atom correspondences.

Table 1 A taxonomy of AAM methods by output objective. Representative methods illustrate the distinction between retaining alternatives and selecting a preferred answer.

	Alternative-generating	**Prediction-oriented**
Objective	Preserve candidate correspondences for later interpretation or selection.	Identify the most plausible correspondence.
Output	Several mappings, stored individually or as compact families.	One preferred mapping.
Examples	SLAPMapper (9); MAPPA (this work).	RXNMapper (14); LocalMapper (1).
Evaluation	Reference coverage: is the annotated mapping retained?	Top-1 accuracy: does the selected mapping match the annotation?

The two objectives can be combined by placing a selector after an alternative-generating method. A selector might favor fewer bond events, apply chemical rules, or use a learned ranking. This separates the ability to discover a useful correspondence from the ability to choose it. Reference coverage measures the former; top-1 accuracy measures the latter. A single predicted mapping can also be evaluated as a candidate set of size one, providing a common coverage criterion across the categories.

For alternative generation, the computational challenge is to retain diverse correspondences without repeatedly searching and analyzing their shared structure. Choices among repeated groups can combine into many complete assignments, even when large parts of those assignments are identical. Enumerating each mapping separately repeats work and obscures which decisions distinguish the alternatives. An efficient representation must preserve those decisions while making their consequences for bond changes accessible. This motivates organizing the search around shared molecular fragments and their allowed rearrangements.

Here we introduce MAPPA, an alternative-generating mapper that treats atom correspondence as a constraint-search problem. Greedy growth discovers fragments compatible with both endpoints; placing one fragment restricts the choices available to the remaining atoms. Molecular symmetry represents rearrangements within these constraints compactly. A matching stage discovers fragment combinations, and a decoding stage relates their alternatives to bond events, reusing calculations across shared fragments. On Golden (10) and EnzymeMap (7), the standard benchmark runs achieve reference coverage of 99.08% and 99.63%, with matching and decoding together averaging 3.19 and 13.16 CPU seconds per reaction, respectively. Broader search raises Golden coverage to 99.57%. We also examine how many alternatives remain under a minimum-event preference, connecting coverage and computational cost to the subsequent task of choosing a mapping.

2 Methods

2.1 Overview: from shared fragments to alternative mappings

For each search direction, the *base* is the endpoint on which fragments grow and the *target* is where their placements are sought. Either endpoint can serve as the base; reversing the search exchanges these roles. MAPPA builds atom mappings by growing structural fragments shared between these endpoints (Figure 1). Starting from a seed atom, it extends a fragment while a compatible placement remains possible. When no admissible extension remains, it finds the completed fragment's alternative placements. Each placement defines a search branch: that choice is locked, and fragment growth continues on the remaining atoms under the accumulated assignments. Repeating this process produces alternative compositions of matched fragments. Different seeds, single-bond cuts and search directions expose additional compositions.

Two stages organize this procedure. The *matching stage* discovers fragments and their conditional placements, then reduces repeated fragments and branch compositions into a shared mapping catalogue. Reduction is its final subphase. The *decoding stage* uses that catalogue to relate permitted atom rearrangements to bond events, preparing each unique fragment once and combining its response with the branch boundaries. Its output retains branches that can be queried within a chosen event-count window, rather than only one selected mapping. Matching and decoding are separate operations; a saved catalogue can be decoded without repeating discovery. Reference verification evaluates these outputs separately (Section 3.2).

2.2 Matching Stage: grow, saturate and branch

Matching alternates between *growth matching*, which tests for a relaxed subgraph isomorphism, and *saturation matching*, which finds the completed fragment's distinct target atom-set placements. Postponing branching lets a small, common motif grow into a larger, more restrictive constraint (Figure 1a).

2.2.1 Growth matching: keep extending while a match exists

The growing fragment $g = (B_g, A_g)$ contains base atoms B_g and accepted edges A_g; $\mathcal{L}$ records earlier locked assignments. A *relaxed subgraph isomorphism* maps these atoms injectively into the target, preserving elements and locks, with each accepted edge placed on a target bond within tolerance τ_{iso}. A *witness* m is

one such map. Explicit hydrogens participate. Matching is non-induced: extra target edges are allowed, and unselected base bonds impose no additional constraint.

Growth starts from a seed with an available target atom. The presence threshold b defines eligible connections. It selects a boundary atom uniformly, then one of its frontier edges uniformly, and tentatively forms g^+. The decision is an existence test:

$$\text{Exists}(g^+ \mid \mathcal{L}) = \mathbf{1}\big[\mathcal{M}(g^+ \mid \mathcal{L}) \neq \varnothing\big] , \tag{1}$$

where $\mathcal{M}$ is the set of relaxed subgraph isomorphisms satisfying the locks. If a witness is found, accept the extension. If no placement exists, defer that edge, keep g, and try the remaining frontier. *Saturation occurs only when no eligible extension remains.* One failed edge does not end growth.

Let $\nu(g)$ count distinct target atom sets admitting a placement (Table 2). Growth does not compute this count: every positive outcome means continue growing, regardless of multiplicity.

Table 2 Match-count transitions during proposed growth. "Many" means more than one target atom-set placement, not multiple symmetry permutations on the same set. The $0 \to 0$ row is a rejected start; a zero-match fragment is never expanded.

$\nu(g) \to \nu(g^+)$	Decision
$0 \to 0$	No seed placement: skip this start and try another seed.
$1 \to 0$	Keep g and defer the edge. At saturation, one placement gives one continuation.
many $\to 0$	Keep g and defer the edge. At saturation, resolve its alternative placements.
$1 \to 1$	Accept the extension and continue growing.
$1 \to$ many	Accept; postpone the placement choice until saturation.
many $\to 1$	Accept; the larger fragment has narrowed the choices.
many $\to$ many	Accept; retain one witness and continue growing.

For $1 \to$ many, consider one eligible target carbon with two oxygen neighbors. C has one target atom set; C–O has two. Finding either placement accepts the extension without branching. The witness can change target sites during later growth, while earlier locks remain fixed. Extending the current witness or reusing a compatible saved fragment can avoid a new search; otherwise the existence solver finds one witness or establishes absence. A resource limit is not a zero-match answer.

2.2.2 Saturation matching: recover distinct atom-set placements

At saturation, MAPPA finds distinct target atom sets $T_i = m(B_g)$ reached by relaxed subgraph isomorphisms. Each pair (B_g, T_i) defines a completed fragment placement, with its internal mapping relation and the binary-graph automorphisms of its two sides retained compactly.

A six-carbon ring fragment fitting two target rings with different external attachments gives two atom-set placements (Figure 3). Ring rotations and reflections are fragment automorphisms: they change the correspondence within a placement without changing its atom sets. Search excludes already represented target sets directly, avoiding enumeration of their bijections followed by deduplication.

Fragment automorphisms act within each endpoint's fragment graph. *Global automorphisms* act on the whole target graph. A certified subgroup of the latter can share work between placements only if it preserves the entire remaining matching problem, moving later fragments jointly and respecting earlier locks. Otherwise those placements remain separate (Appendix A.3).

2.2.3 Hierarchical branching: lock a placement and continue

Each completed placement creates a child continuation: lock its assignments and grow the next fragment. In Figure 1a, choosing A$\to a$ leaves sites b, c for B; choosing B$\to b$ leaves c for C. These are conditional decisions: two fragments cannot independently occupy the same target atoms. Joining an earlier fragment retains its locked assignments and accepted constraints.

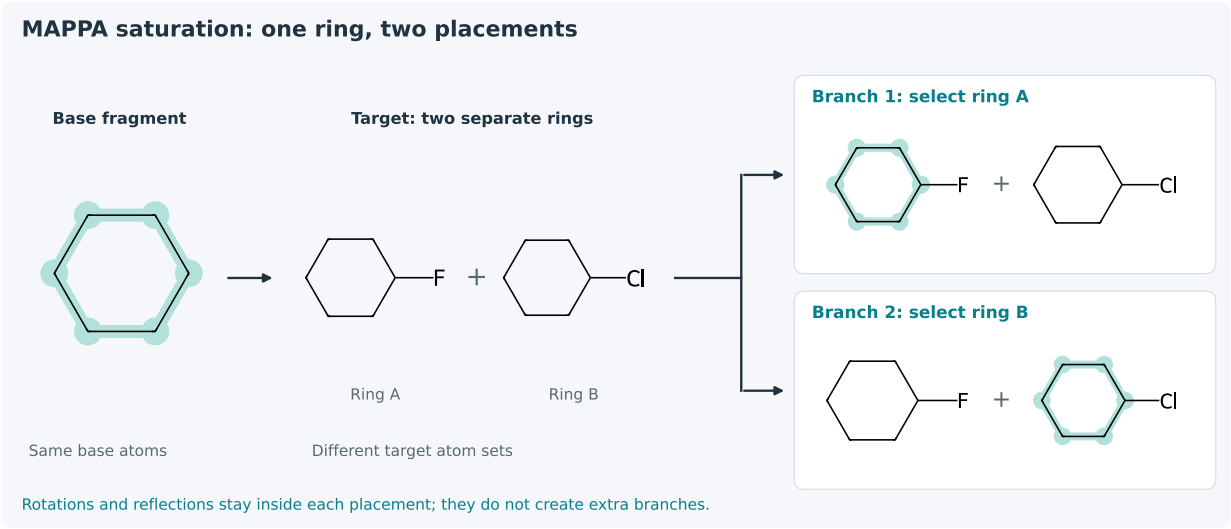


Figure 3 Two ring placements give two branches. A fixed six-carbon fragment fits either target ring. The different F and Cl attachments prevent a global automorphism from exchanging the rings. Each child selects one target carbon atom set, highlighted in teal; rotations and reflections within that ring remain inside the placement rather than creating additional branches. This is a carbon-fragment illustration with hydrogen constraints omitted, not a complete reaction.

Each locked fragment expresses a shared structural constraint, extended as far as the current greedy trajectory permits. Another seed may discover a different constraint. After each completed step, apply the branch cap from leaves to root: discard the largest sibling subtrees as needed, recount, then assess the parent. Internal symmetry permutations do not count as branches. Caps restrict coverage and are recorded explicitly (Appendix A.4).

2.3 Matching Stage: seeds, cuts and search direction

Vary the starting point. One greedy trajectory can repeatedly favor a large preserved region and miss another useful decomposition. In Figure 4a, the highlighted benzyl region offers more starting atoms than the remaining methylamine region. Additional seeds can discover different constraints, but may also revisit the same large fragment.

Vary the allowed connections. A single-edge *cut sweep* perturbs this commitment systematically. MAPPA searches the uncut graph and then repeats the search with each eligible base edge temporarily masked, including bonds to hydrogen. Each cut condition receives the requested number of seed orderings. Masking a connection inside the dominant region can expose a trajectory that ordinary seeding rarely reaches. The original endpoint weights remain unchanged for decoding: a mask alters discovery and does not assert a broken bond. Its endpoints may reconnect through another route, and one cut can redirect several later fragment choices.

Exchange the endpoints. Exchanging the base and target adds further alternatives. Seeds, cuts and reverse searches broaden discovery without making it exhaustive. The same exploration principle also broadens SLAP's returned alternatives: the expanded protocol described in Section 4.2 reaches approximately 97% coverage.

2.4 Matching Stage: consolidate unique fragments and branches

Different seeds and cuts often rediscover the same fragment. The final matching subphase, *reduction*, gives each base/target atom-set pair (B_i, T_i) one shared identity F_i. For example, two searches that match $\{1, 2, 3\}$ to $\{a, b, c\}$ reference the same F_i, even if their growth order or starting witness differs. Each branch then references its fragment identities, allowing repeated compositions to share one catalogue entry.

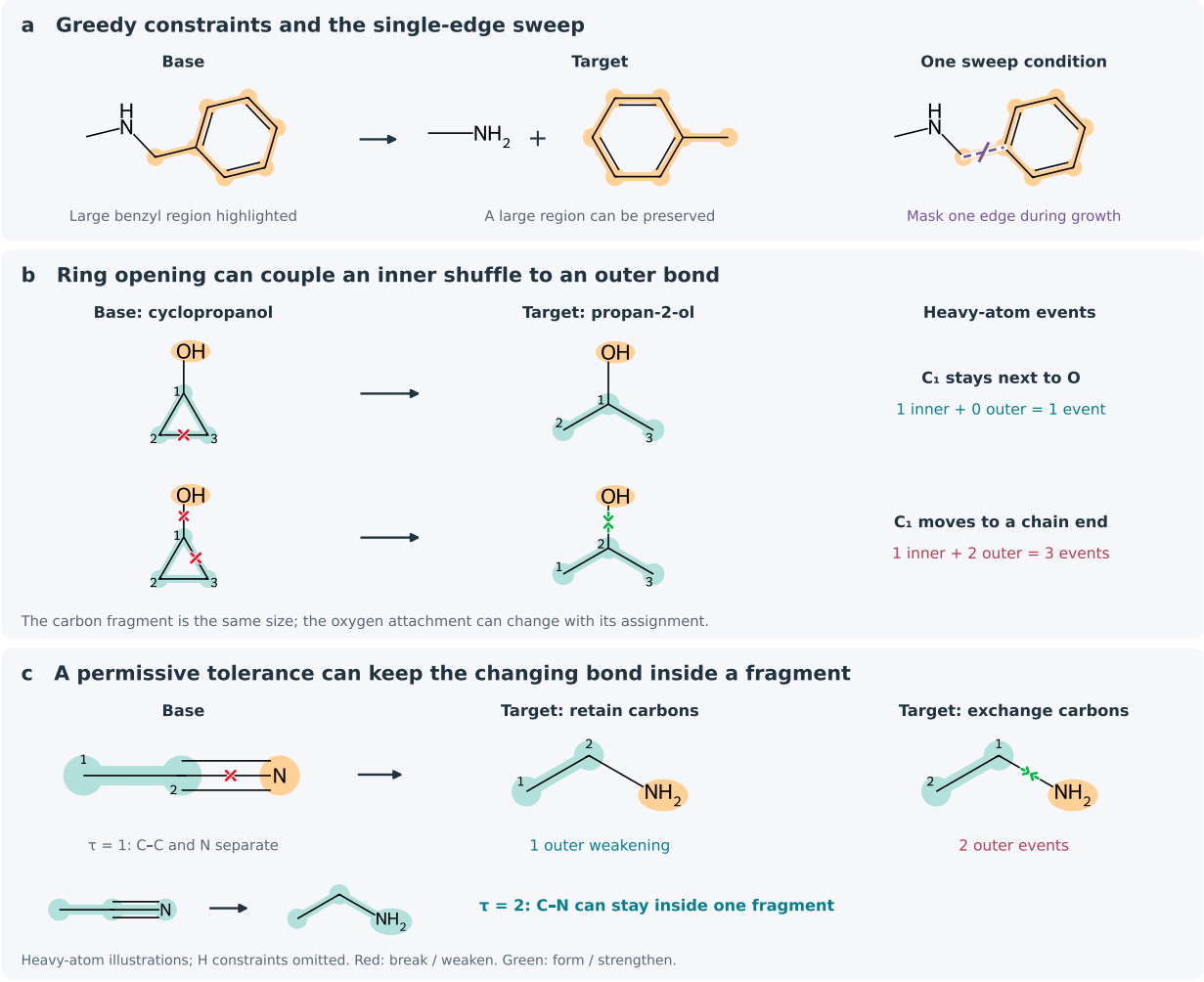


Figure 4 Molecular examples of exploration and boundary dependence. (a) The benzyl region is preserved between N-methylbenzylamine and the displayed methylamine/toluene endpoint. Masking one of its edges changes the growth conditions; the purple mark is a search mask, not a claimed cleavage. (b) In cyclopropanol and propan-2-ol, the teal carbon fragment has one internal ring-opening event under either displayed assignment. Carbon 1 at the chain center preserves its oxygen attachment; at an end, it adds an outer cleavage and formation. The heavy-atom totals are one and three. (c) For acetonitrile and ethylamine, tolerance one can leave C–C and N separate: exchanging the carbons changes one outer weakening into two outer events. Tolerance two can include the changing C–N bond inside one fragment. Numbers track the same carbon atoms; red and green marks indicate losses and gains as in Figure 1. These are heavy-atom graph illustrations, with hydrogen constraints omitted and event threshold 0.5; panels b and c are checked by direct pair counting. They illustrate possible boundary dependence, not its measured frequency.

A fragment identity is a key, not permission to map its atoms arbitrarily. Reduction keeps one original record with its witness, discovery constraints, anchors and symmetry relation. Branches also retain the ordered placement context needed to move later fragments consistently. Equal compositions are consolidated with that context preserved; their cut provenance is collected together. Selecting one original record does not retain the union of all discarded discovery relations.

This subphase reuses the fragment inventories already accumulated during matching. It performs structural grouping, with no symmetry optimization or event computation. Its output is one shared catalogue per reaction and direction. That catalogue is the input to decoding.

2.5 Decoding Stage: prepare each fragment's symmetry–event response

Decoding turns the retained mapping alternatives into queryable bond-event information. It first prepares each unique fragment once, then reuses that response wherever the fragment occurs. The local calculation connects *which atoms can move together* to *which bond changes those moves produce*.

2.5.1 Fragment automorphisms: independent and correlated moves

A *fragment automorphism* permutes atoms within one endpoint's fragment while preserving its binary connectivity and element labels, with anchors fixed. We compute these automorphisms separately for the base and target sides. Bonds satisfying the presence criterion share one structural label; original bond weights are retained for event counting. External attachments do not constrain the fragment automorphisms.

Nauty supplies compact generators of these automorphism groups (11): unit operations that describe the allowed rearrangements. Exchanging two identical arms moves each arm with its attached atoms; exchanging two hydrogens can act within an arm. The two arms must still occupy different positions. These joint operations retain that correlation without storing every full permutation. Both endpoint actions are represented; for an aligned equal-connectivity pair, one side suffices to describe their relative rearrangements.

2.5.2 Internal events: attach bond changes to the allowed moves

Let W_B and W_T denote the original base and target bond-weight matrices. For mapped atoms u, v, an assignment m changes the bond weight by

$$\Delta W_{uv}(m) = W_T(m(u), m(v)) - W_B(u, v). \tag{2}$$

The pair contributes one event when $|\Delta W_{uv}|$ exceeds its event threshold. We use empirical bond-order-change thresholds of 0.5 for organic bonds (the default for non-metal pairs) and 0.3 for bonds involving at least one metal atom. The sign distinguishes formation or strengthening from cleavage or weakening. These thresholds differ from the growth tolerance: a binary symmetry can preserve connectivity while changing bond weights.

In the acetate example (Figure 1c), O3 and O4 are interchangeable under an automorphism of the fragment's binary connectivity: each oxygen is adjacent to the same carbon. Their original C–O bond orders differ, so exchanging their assignments adds one weakening and one strengthening, giving two internal events. The saved response associates the rearrangement with those changes, so every branch using that fragment can reuse the calculation. Independent symmetry components are processed separately when no event term or requested endpoint couples them. Event-invariant choices share computation; correlated choices remain joint. The response supports later event queries without expanding every complete mapping.

2.5.3 Different connectivity: improve the starting correspondence

The accepted growth graph can omit bonds present in the final fragment. In the substituted triangle–chain example (Figure 4b), the displayed local shuffles keep one internal ring-opening event while changing the outer attachment. More generally, two valid relaxed subgraph isomorphisms on the same atom sets need not be connected by the final fragment automorphisms. Optimizing only one starting correspondence can therefore miss a lower-event alternative.

MAPPA first improves the saved witness under fragment automorphisms. Additional starting-correspondence search is allowed only when growth and final connectivity differ on at least one side and the witness has not

attained an established event lower bound. Preparation and improvement share a fixed per-fragment budget. The output retains the selected starting correspondence and its symmetry response. An unsettled minimum is recorded as best found; selecting this starting orbit restricts decoded coverage. Solver details and bounds are given in Appendix A.6.

2.6 Decoding Stage: assemble branch events and select the window

A branch combines three contributions: internal fragment events, ordinary boundary events and the correction for the swept bond. For a branch $\mathcal{B}$ and a permitted correspondence m, the reported count is

$$E_{\mathcal{B}}(m) = B_{\mathcal{B}} + \sum_i e_i(m) + s_{\mathcal{B}}(m), \qquad 0 \le s_{\mathcal{B}}(m) \le 2. \tag{3}$$

Here e_i is the reused internal response of a disjoint final fragment piece, $B_{\mathcal{B}}$ is the ordinary-boundary contribution, and $s_{\mathcal{B}}$ accounts for the swept connection. The following rules specify how they are combined.

2.6.1 Ordinary boundaries: a fixed contribution under the model

MAPPA uses the independent ordinary-boundary approximation: these contributions remain fixed as internal fragment choices change. Bonds touching mapped atoms are counted, including mapped/unmapped boundaries; bonds wholly within unmatched material are omitted. No assignment is invented for an unmatched atom.

Greedy growth helps absorb connections that would otherwise couple fragments. In C–C≡N → C–C–N (Figure 4c), tolerance one can leave C–C separate from N. Swapping the carbons changes the outer contribution from one weakening to a cleavage plus a formation. Tolerance two allows C–N inside the fragment, where the internal response handles that dependence. Greediness helps, but does not guarantee independence: the triangle–chain example still has a variable outer attachment. Equation (3) is therefore a model count, which can differ from the physical whole-reaction count.

Random growth, multiple seeds and the cut sweep can mitigate this limitation by discovering alternative fragment decompositions. In Figure 4b, growing through the O–C attachment first can place it inside a shared fragment instead of leaving it across a boundary. Another branch can then represent the same correspondence with its correct event count, even if the original branch assigns it differently. This exploration does not repair the original branch's approximation or guarantee a correct representation elsewhere. The benchmarks do not isolate the size of this mitigation.

2.6.2 The swept bond: couple neighboring fragment responses

A sweep masks one base bond. When its endpoints lie in different final fragments, those fragments are neighbors across the masked connection. Their internal event responses can be prepared separately, but this boundary contribution depends jointly on their choices: preserving the connection requires the two atoms to occupy compatible target positions. The swept bond therefore introduces a coupling between neighboring fragments that the decoder handles explicitly.

During branch assembly, MAPPA reuses the prepared responses to evaluate the two endpoint positions together, retaining their dependence on internal events and one-to-one occupancy. If both endpoints remain in one fragment, the same query uses that fragment's response. The associated contribution $s_{\mathcal{B}}$ ranges from zero to two events: a correspondence can preserve the connection, produce one bond change, or break one connection and form another. The sweep still masks only one bond; the event count describes the resulting correspondence across the two molecular endpoints.

2.6.3 The event window: retain mappings near the minimum

The event window focuses decoding on mappings with few bond changes. MAPPA first adds the local fragment minima and fixed boundary contribution for each branch. Let L be the smallest of these totals before resolving the swept bond. When the fragment minima can be combined, the sweep adds at most two events, giving an initial branch count no greater than $L + 2$. Only branches whose lower bounds are at

most $L + 2$ need to be examined for a lower count. Their swept-bond contributions are evaluated using the prepared fragment responses.

The lowest count found, U, sets the requested window $[U, U + n]$. MAPPA retains branches that could contribute mappings within this window and excludes any branch whose lower bound exceeds $U + n$. The studies here use $n = 5$. Retained branches reference the shared fragment responses, allowing subsequent queries for mappings at a chosen event count or with specified atom correspondences.

2.7 Execution: parallel stages and shared computation

The stage boundaries also define parallel work. Reactions and directions are independent; seed/cut searches within a direction can run concurrently before their results are reduced together. Growth along one branch remains sequential because later fragments depend on earlier occupancy. For the reported batch workflow, the matching stage, including reduction, finishes before catalogue decoding begins. Different reaction/direction catalogues can then be decoded independently, with unique-fragment responses shared within each catalogue.

A persistent worker pool schedules these operations through the same interfaces used by direct local calls. Intermediate records can remain in memory or use aggregated storage. Compatible fragment proofs can be reused during growth, but each seed and cut keeps its own random decisions. Matching and decoding use a shared symmetry core, and bulk operations use native kernels. Scheduling and persistence do not determine which mappings are allowed. Detailed solver, cache and storage protocols, together with full pseudocode, appear in Appendix A.

3 Evaluation

3.1 Datasets and search settings

We evaluate all 1,851 Golden reactions (10) and the prepared collection of 47,974 EnzymeMap reactions (7). Reference labels are removed before search. Both studies use the prepared formal-bond-order matrices and explicit hydrogens. EnzymeMap supplies curated database mappings as references.

The common settings are discovery tolerance $\tau_{\text{iso}} = 2$, presence/cut threshold $b = 0.2$, hierarchical branch cap 100, random seed 42, both directions and the full single-edge sweep. Golden uses one and ten seed orderings per cut; EnzymeMap uses one. Decoding uses the original bond orders and empirical event thresholds of 0.5 for organic bonds and 0.3 for bonds involving a metal, with a five-second allowance per unique fragment and the best attained model count plus five. No reference mapping guides growth, pruning or starting-correspondence selection.

We compare one-seed MAPPA with two SLAPMapper configurations on identical Golden inputs and the same CPU model. Standard SLAP uses the uncut binary graph; expanded SLAP combines binary and weighted modes with the full single-edge cut sweep. All three configurations run in both directions and use the same chemical endpoint-equivalence rule for reference verification.

3.2 Catalogue and decoded-reference verification

We report two external coverage measurements. *Catalogue recovery* asks whether the reference belongs to the retained matching relation. *Decoded recovery* requires an actual injective mapping from a saved starting correspondence and its joint symmetry choices, in a retained branch and event window. A positive certificate is replayed and its event count independently checked. A catalogue certificate may suggest a candidate, but cannot alone establish decoded membership.

The decoded audits preserve exact annotated heavy-atom indices and unmatched-heavy-atom requirements. Unannotated hydrogens must admit a feasible completion. Golden catalogue verification additionally permits endpoint chemical symmetries preserving charge, isotope, hydrogen count and stereochemistry; the Golden catalogue and decoded scores therefore differ in both representation and reference-equivalence policy. EnzymeMap uses exact indices for both. Recovery means inclusion in either search direction, not top-ranked

prediction accuracy. Every reaction remains in the denominator, and unresolved queries remain separate from absence.

3.3 Timing and reference-event analysis

Matching time includes both fragment discovery and reduction into the final catalogue; decoding is reported separately. CPU seconds sum successful worker computation over both directions and all cuts, including stage input loading and result construction. Publication and external verification are excluded. Source identities, hardware and completed-pass accounting are documented in the supplement.

To locate the reference relative to the detected minimum, let m_{GT} be a saved, verified mapping that agrees with the annotated atom pairs. We measure

$$n_{\mathrm{GT}} = E(m_{\mathrm{GT}}) - U, \tag{4}$$

where U is the detected minimum already saved by decoding, taking the lower count across both directions. When both directions supply verified mappings, we use the one with the lower recorded count. Both terms use the same all-atom decoder model. Unannotated atoms, including hydrogens, retain their assignments in the verified witness. This comparison uses existing counts without further optimization.

For references absent from the saved output, we fix their annotated pairs as anchors and complete the unspecified assignments with one seed per direction and no cut sweep. We compare the lowest attained completion count across both directions with the original U. The supplied pairs remain fixed, and the completed mapping is checked against the original bond matrices. These completion offsets are added to the same distribution, so that every reaction contributes once. Anchored completions do not contribute to independent mapping coverage.

We separately count the distinct heavy-atom mapping classes actually attainable at the detected minimum, taking the union across retained branches and both directions. Chemical endpoint equivalence identifies repeated representations, and hydrogen permutations do not multiply the choice count; each class must admit a full valid completion. The ranking score remains the saved all-atom model count. If a reaction has K choices and its reference class is present, a uniform selector succeeds with probability $1/K$; otherwise it succeeds with probability zero. This defines expected random-choice accuracy without a trained selector. Incomplete counts contribute bounds, not exact values. This analysis tests for any feasible hydrogen completion at the minimum; Equation (4) instead uses the recorded reference witness.

4 Results

4.1 Reference coverage across datasets and methods

MAPPA recovers 1,834 of 1,851 Golden references with one seed (99.082%) and 1,843 with ten seeds (99.568%). On EnzymeMap, one seed recovers 47,798 of 47,974 references (99.633%; Table 3 and figure 5). Coverage measures reference inclusion among the returned alternatives. Matching, decoding and reference evaluation are complete for both datasets.

Table 3 Reference coverage on Golden and EnzymeMap. Catalogue coverage measures the complete matching output; decoded coverage measures the saved minimum-plus-five event window. Golden catalogue evaluation permits chemical endpoint equivalence, whereas decoded evaluation uses exact annotated indices. EnzymeMap uses exact indices for both. Percentages use all reactions.

Study	Reactions	Catalogue	Decoded	Decoded absent
Golden, 1 seed	1,851	1,834 (99.082%)	1,833 (99.028%)	18
Golden, 10 seeds	1,851	1,843 (99.568%)	1,842 (99.514%)	9
EnzymeMap, 1 seed	47,974	47,798 (99.633%)	47,796 (99.629%)	178

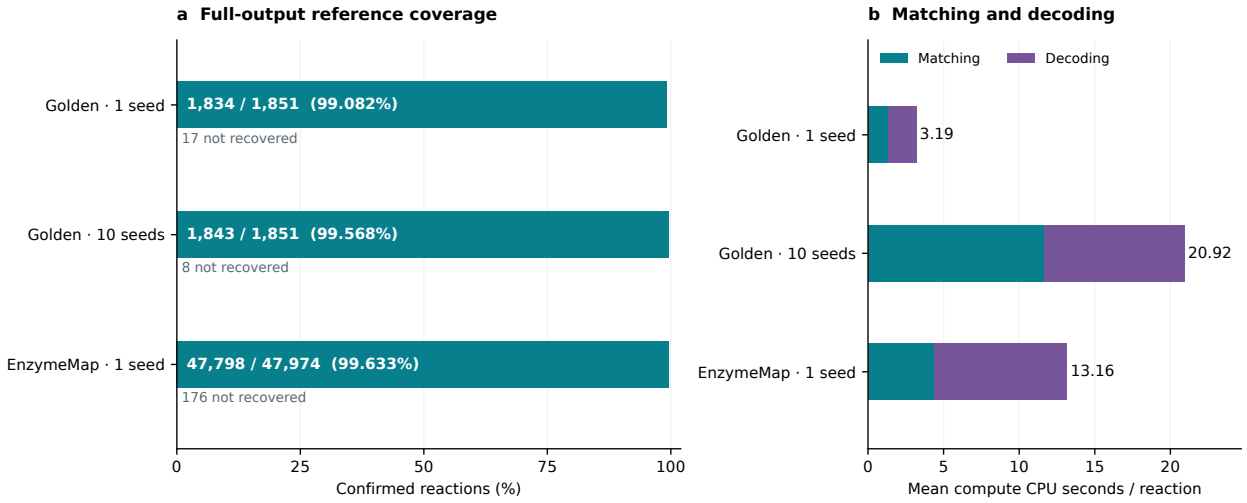


Figure 5 Reference coverage and computational cost. (a) Reference inclusion in the complete matching output. (b) Mean CPU seconds per reaction for matching, including reduction, and decoding. Both directions and the full cut sweep contribute; publication and external verification are excluded.

The decoded minimum-plus-five window retains 1,833 Golden references with one seed, 1,842 with ten, and 47,796 EnzymeMap references. Ten seeds recover nine additional Golden references without losing any recovered by one seed. The wider exploration therefore improves coverage even when the same decoding window is retained.

Table 4 compares Golden coverage across methods. A single prediction is treated as a candidate set of size one. MAPPA achieves the highest coverage among the evaluated methods. LocalMapper has the highest coverage among the evaluated single-answer interfaces at 86.71%, followed by Chython at 84.33% and RXNMapper at 83.58% (1; 2; 14). Default binary SLAP covers 85.95%; combining binary and weighted modes with both search directions and the cut sweep raises its coverage to 96.92%. These results show the value of broader correspondence discovery before selecting one answer.

Table 4 Golden reference coverage across AAM methods. All methods are evaluated over 1,851 reactions using chemical endpoint equivalence. Coverage counts reference inclusion anywhere in the returned output; a single prediction is a singleton set. MAPPA uses the full one- and ten-seed studies. Expanded SLAP includes both weight modes, both directions and the cut sweep.

Method	Search setting	Output	Coverage
MAPPA	1 seed, sweep	Families	1,834 (99.08%)
MAPPA	10 seeds, sweep	Families	1,843 (99.57%)
SLAP	Expanded sweep	Candidates	1,794 (96.92%)
SLAP	Binary, both directions	Candidates	1,620 (87.52%)
SLAP	Binary, default	Candidates	1,591 (85.95%)
LocalMapper	Default	Single mapping	1,605 (86.71%)
Chython	Default	Single mapping	1,561 (84.33%)
RXNMapper	Default	Single mapping	1,547 (83.58%)
RDT	Default	Single mapping	1,030 (55.65%)
Indigo	Default	Single mapping	692 (37.39%)

4.2 Computational cost and seed scaling

One-seed MAPPA requires an average of 3.19 CPU seconds per Golden reaction: 1.31 for matching and 1.88 for decoding. EnzymeMap requires 13.16 CPU seconds, divided into 4.41 for matching and 8.75 for decoding (Table 5). The EnzymeMap median is 5.80 seconds and its 95th percentile is 41.86 seconds, indicating that a minority of difficult reactions account for much of the total computation.

Table 5 Mean CPU seconds per reaction. Matching includes fragment discovery and reduction. Both directions and the full cut sweep are included. Decoding is measured separately; publication and external verification are excluded.

Study	Matching	Decoding	Total
Golden, 1 seed	1.31	1.88	3.19
Golden, 10 seeds	11.64	9.28	20.92
EnzymeMap, 1 seed	4.41	8.75	13.16

Increasing Golden from one to ten seeds raises matching cost by 8.87-fold and decoding cost by 4.95-fold. Matching scales approximately linearly with seed count over this range. Decoding grows less because repeated fragments and branch compositions share their event responses. The additional exploration recovers nine more references at an average total cost of 20.92 CPU seconds per reaction.

The paired SLAP comparison illustrates the coverage–cost trade-off (Figure 6). Standard binary SLAP is fastest. MAPPA achieves higher coverage than expanded SLAP while requiring less computation, including its separate decoding stage. Detailed measurement scopes are given in Appendix B.

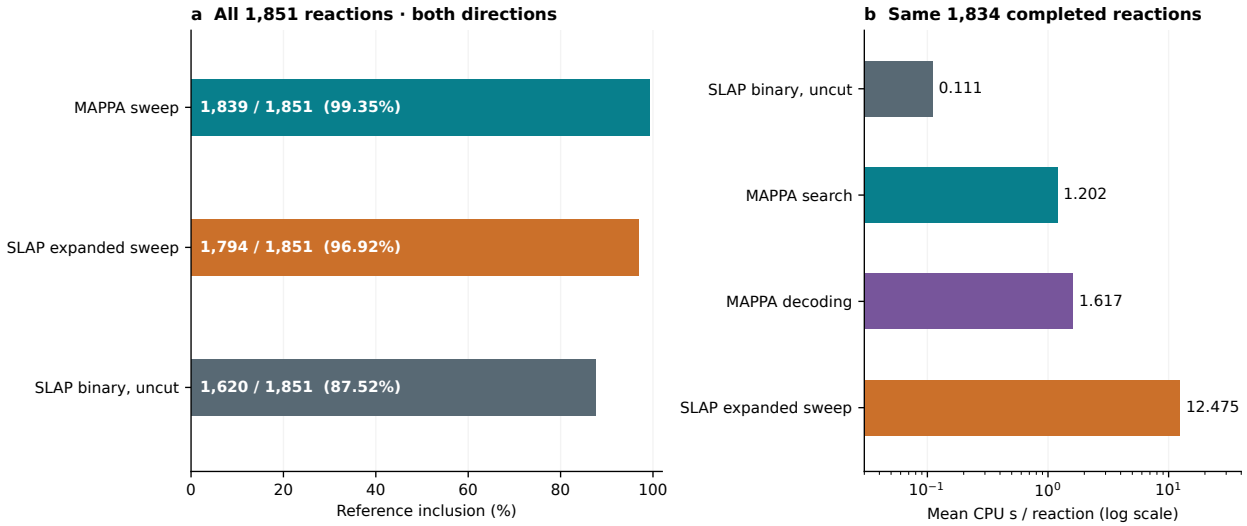


Figure 6 Coverage and cost under paired evaluation. (a) Reference coverage over all 1,851 Golden reactions in the paired study. Standard SLAP is binary and uncut; expanded SLAP combines binary and weighted modes with the cut sweep. All protocols use both directions. (b) Mean CPU time on the 1,834 reactions completed by all protocols, measured on the same CPU model. MAPPA search and decoding are separate; the time axis is logarithmic.

4.3 Reference event counts relative to the detected minimum

The saved GT-consistent mappings have the same event count as the detected minimum for 1,630 Golden reactions (88.06%) and 44,086 EnzymeMap reactions (91.90%; Figure 7). Most other mappings lie within five events above the minimum. These distributions show that low event counts often agree with the reference, while retaining nearby alternatives provides additional reference-consistent choices.

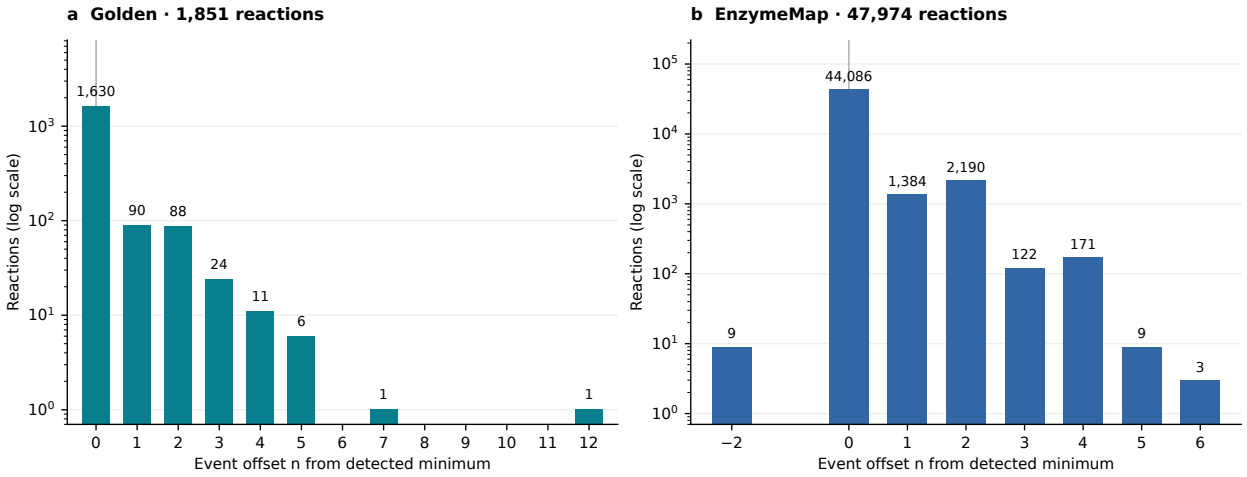


Figure 7 Reference-mapping event counts relative to the detected minimum. Each reaction contributes $n = E(m_{\mathrm{GT}}) - U$, using a verified GT-consistent full mapping and the original saved detected minimum. The lowest recorded verified count across both directions is used, with no additional optimization. Saved anchored completions supply the 18 Golden and 178 EnzymeMap references absent from the original output. Both terms use the same all-atom decoder model. Labels give reactions per bin; the vertical axis is logarithmic. The nine negative EnzymeMap offsets are anchored completions two events below the original detected minimum. Independent coverage is unchanged.

Each offset describes a concrete reference-consistent mapping, including its saved hydrogen completion. It does not minimize over alternative completions of the reference.

Anchored completion succeeds for all 196 previously absent references, included in Figure 7. The completed count equals the original detected minimum for nine Golden and 83 EnzymeMap cases; nine EnzymeMap completions lie two events below it. The remaining completions lie above the detected minimum, reaching offsets of 12 in Golden and six in EnzymeMap. Every supplied pair is preserved, and each selected completion's model count agrees with direct counting from the original bond matrices.

4.4 How many choices remain at the minimum?

Minimum-event selection produces a compact shortlist (Figure 8). Among 1,829 Golden reactions with exact choice counts, 1,370 have one choice, 352 have two, 96 have three to five, and 11 have six or more. Their mean is 1.36 choices. Across all 1,851 reactions, the median is one and the 95th percentile is three, including the bounds on the remaining counts.

The reference belongs to the minimum-event set in 91.36–91.46% of Golden reactions. Selecting uniformly among the available minimum-event choices gives an expected accuracy of 78.31–79.45%. For example, Golden case 1694 has eight minimum-event choices, including the reference, so a random selector succeeds with probability 1/8. MAPPA exposes these alternatives for subsequent chemical selection; a reference outside the minimum set requires considering a wider event window.

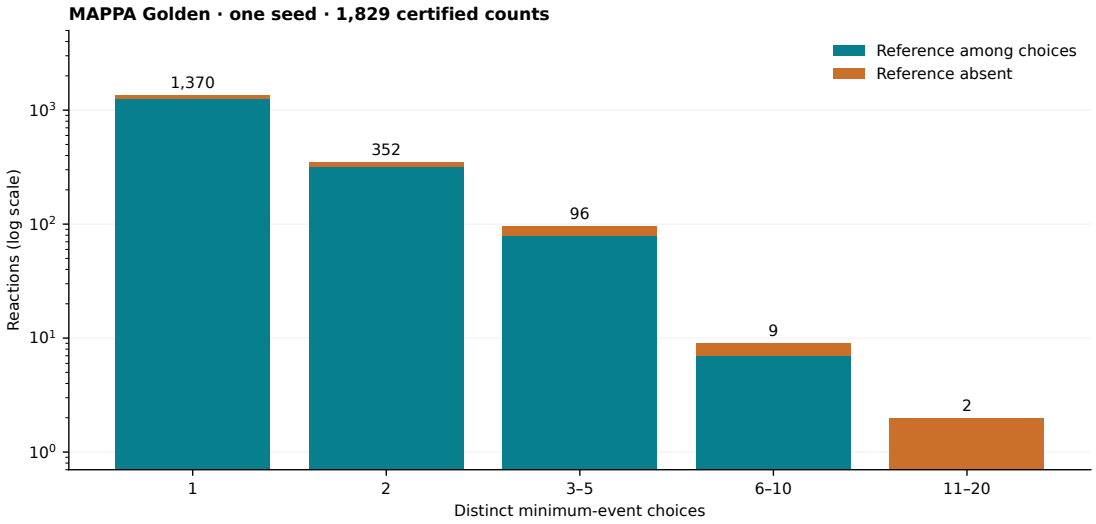


Figure 8 Distinct choices under minimum-event selection. Bars cover the 1,829 one-seed Golden reactions with exact counts; the remaining 22 counts contribute bounds to the full-dataset statistics. Equivalent heavy-atom relabellings and hydrogen permutations count once. Colors indicate whether the reference belongs to the minimum-event choices. The vertical axis is logarithmic.

5 Discussion

MAPPA combines high reference coverage with explicit retention of mapping alternatives. Its fragment representation organizes these alternatives around shared structural constraints, allowing the same local relationship to contribute to many branches. Matching discovers those constraints, and decoding connects their permitted rearrangements to bond changes. This separation supports both the generation of candidate correspondences and the interpretation of how those candidates differ.

The seed study illustrates a practical trade-off between exploration and computation. One seed recovers most references, while additional seeds provide a modest coverage gain with approximately proportional matching cost. The cut sweep offers a complementary way to vary the constraints encountered during greedy growth. Together, these controls allow search effort to be adjusted to the application, with independent searches available for parallel execution.

The event distributions suggest that a small set of low-event alternatives can provide a useful starting point for chemical selection. In Golden, the median number of distinct minimum-event choices is one, although some reactions retain several alternatives. Bond-event count can therefore narrow the choices before reaction-specific knowledge is used to distinguish equally economical assignments or favor a higher-event solution. It remains a surrogate for chemical plausibility, and the boundary approximation described in Methods affects the interpretation of individual counts.

Retained alternatives could also assist reaction-path preparation. Atom correspondence determines how endpoint coordinates are paired during interpolation (3); different assignments can therefore supply distinct initial paths for methods such as the nudged elastic band (8). Coupling these alternatives to chemical ranking and path refinement could help select correspondences using information beyond endpoint structure. Beyond pathway preparation, MAPPA could support reaction-template extraction from mapped bond changes. It could also serve as a structural verifier during reinforcement-learning training of large language models, providing feedback on proposed atom correspondences and bond changes.

Code and data

Code and reproduction instructions are available at MAPPA source repository, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden and EnzymeMap are distributed by their original authors (10; 7).

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A Technical algorithm details

The complete stage sequence is summarized in Algorithm A1. The following sections give the constraints and implementation details underlying the main-text description.

Algorithm A1 Matching (including reduction), followed by decoding

Require: Endpoint graphs, discovery policies, seed/cut plan, branch cap and event window
for all requested search directions **do**
  for all cut conditions and seed orderings **do**
    Initialize the conditional search hierarchy
    while a retained continuation has an eligible fragment seed **do**
      Grow using one-witness existence queries
      At saturation, construct distinct compressed placements
      Continue each placement with earlier assignments locked
      Prune and recount the hierarchy from leaves to root
    end while
    Retain terminal compositions and compact provenance
  end for
  Share fragments by atom-set pair; reduce repeated branch compositions
end for
Complete matching, including reduction, before batch decoding
for all reaction/direction catalogues **do**
  Inventory unique fragments and prepare their shared symmetry-event responses
  Compose branch contributions and the sweep-endpoint correction
  Retain queryable branches within the requested event window
end for

A.1 Introductory alternatives example

The EnzymeMap case 13175 witnesses in Figure 2 come from the saved one-seed benchmark with branch cap 100, discovery tolerance two, the full single-edge sweep and a decoded minimum-plus-five window. Matching and decoding were not rerun for the figure. A is the complete witness retained by the exact-index decoded-reference audit; all nine annotated heavy-atom pairs agree. B was queried from the saved response at its detected minimum of six all-atom events and replayed through its saved symmetry choices. Both saved witnesses map all 15 atoms injectively. Direct counts from the original endpoint matrices give eight and six all-atom events, respectively, agreeing with their model counts. Their four and two heavy-atom events are shown in the figure. Events require an absolute bond-weight change greater than 0.5.

Because EnzymeMap does not annotate hydrogen identities, we separately checked all $6! = 720$ hydrogen assignments for each fixed heavy-atom mapping. Both permit a minimum of six total events. For A, transferring the carboxyl OH hydrogen to the formaldehyde oxygen contributes two events; the other hydrogen bonds can remain intact. For B, transferring both formaldehyde hydrogens from carbon 1 to carbon 4 contributes four events. The saved A witness contains an additional hydrogen exchange and is not the best hydrogen completion. This independent check does not assert that the newly completed hydrogen assignments belong to the saved decoded response, nor that six is a global minimum over other heavy-atom correspondences. The original saved witnesses remain unchanged. The example therefore illustrates distinct atom origins with tied total counts after hydrogen completion. Portable evidence preserves the endpoint matrices, saved mappings, fragment partitions, replay choices, source checksums and hydrogen-completion check in `enzymemap_case13175.json`.

A.2 Exact existence queries during growth

An existence query seeks a relaxed subgraph isomorphism for the proposed base atoms and accepted preservation edges, subject to completed-fragment locks. Element-compatible destination domains exclude targets already occupied by other fragments. Each accepted edge supplies a binary support relation satisfying target

bond presence and the discovery tolerance. This matching is non-induced: extra target edges are allowed; base nonedges and unselected molecular edges do not add constraints.

More formally, let V_B, V_T denote the base and target atom sets, Z_B, Z_T their elements and W_B, W_T their original bond-weight matrices. A partial correspondence $m : D \subseteq V_B \to V_T$ is injective. For every accepted preservation edge $(u, v) \in A$, its assignment satisfies

$$Z_B(u) = Z_T(m(u)), \quad W_T(m(u), m(v)) \geq b, \quad |W_B(u, v) - W_T(m(u), m(v))| \leq \tau_{\text{iso}}. \tag{5}$$

Here b is the graph-presence threshold and τ_{iso} is the discovery tolerance. These constraints apply to the selected growth graph, not every bond in the completed atom set.

Propagation removes occupied singleton destinations and destinations without support in a neighbor's domain. An empty domain or incompatible singletons establish infeasibility. A bipartite distinct-target check can reject domains that cannot accommodate an injective assignment. These are necessary conditions; surviving ambiguous cases require search. The solver chooses a constrained atom, propagates its tentative destination and stops at the first complete witness. Proving absence can require exploring every remaining feasible choice. Positive and negative answers use the same constraints.

Direct witness extension and indexed fragment reuse precede this query. They establish existence only. Neither supplies a different continuation policy nor converts a timeout into absence. Completed-fragment assignments remain fixed while assignments internal to the current growing fragment may change.

The molecular overview in Figure 1 uses constructed alcohol placements in panel a and methyl acetate saponification in panels b and c. Hydrogens participate in matching but are implicit in the drawings. Exchanging the acetate oxygen images adds two internal changes to two boundary changes at event threshold 0.5. Black lines show endpoint bonds. These illustrations are not a complete search trace, and their formal-bond-order signatures are not distinct mechanisms or resonance-normalized transformations.

A.3 Completion, fragment automorphisms and global placement

Completion searches the finished fragment constraints for distinct target atom-set placements. It excludes already represented sets or certified global placement orbits within the search. It does not enumerate all internal assignments before deduplicating them. Target sites that cease to be globally equivalent remain separate placements, even when their local matched subgraphs share symmetry.

For the fixed carbon-ring illustration in Figure 3, an independent substructure check finds exactly two target atom sets, each admitting twelve ring rotations and reflections. The F and Cl attachments prevent whole-target exchange of those sets. Hydrogen constraints are omitted from this illustration. The check establishes the depicted placements; it is not a full reaction benchmark or the production search procedure.

For each side of a fragment, nauty receives an element-colored binary graph with individually fixed anchors and returns generators of its automorphism group. These compact generators retain correlated moves of rings, arms and their attached atoms. Continuous weights do not label this graph; they remain in discovery and event predicates. Permitted local rearrangements are compositions of these automorphism generators. An accepted relaxed subgraph isomorphism can be followed by base and target fragment automorphisms without reimposing discovery tolerance on the rearranged result. The event response evaluates the resulting original-weight changes.

The subgroup of global target automorphisms used to compress later continuations is more restrictive. It must preserve the full target weight matrix, active connectivity, current locks and every base-to-target compatibility predicate relevant to subsequent growth. Such an automorphism moves the whole later continuation, preserving occupied targets and its matching problem. A fragment automorphism cannot be substituted for this global equivalence proof. Saved placement operations retain their order; unrestricted closure of an ordered sequence would permit combinations not represented by the search.

A.4 Hierarchical caps and provenance

Each retained placement creates a child continuation. After a completed step, count surviving continuations bottom-up. At an over-limit node, discard its largest child subtree, recount and repeat; ties retain the

earlier decision. An ancestor then sees the reduced count. The completed-fragment limit separately rejects a placement expansion exceeding its allowance. Neither operation counts internal symmetry assignments as separate branches, and both retain explicit cap evidence.

A shared search state may have several incoming histories. The retained graph keeps parent links and terminal references rather than unfolding every path. States and payloads with no surviving descendant are released. Returned histories and cut provenance remain available for inspection, but are not independent fragment-decoding jobs. A branch cap bounds retained choices, not every temporary allocation or total memory.

A.5 Fragment identities and composition reduction

Reduction reuses the fragment inventory in each search graph and unifies it across seeds and cuts. A completed fragment's key is its base/target atom-set pair within a fixed endpoint problem and policy. Keep the first original record intact. Other starting witnesses, growth routes, domains or anchors do not create additional identities and are not unioned into broader permissions. Raw archives retain the original decisions; the reduced catalogue retains the selected record and compact provenance. When reducing an explicitly selected path subset, the representative must come from that subset.

Native projection converts retained states into shared fragment identifiers. Composition keys use those identifiers together with the ordered placement context. Independent decisions can commute; overlapping local or global choices retain their order. Equal keys share one family record and aggregate cut/partition provenance. Reduction uses structural equality, not symmetry solving or event queries. The atom-set key must not be confused with a permission to map its atoms arbitrarily.

A.6 Shared event core and bounded starting-map improvement

Each unique fragment is prepared once. For a selected starting correspondence, the shared core stores binary symmetry generators, fixed event contributions and sparse variable original-weight terms. It separates operation components only when no variable event or requested endpoint joins them. A generator moving two arms couples those moves even if each arm also has its own independent internal exchanges. Event-preserving operations can share computation only when they also preserve all endpoints relevant to the query. No complete group or Cartesian table of mappings is stored.

Starting-map improvement first compares the saved witness with lower bounds. One bound matches pair-weight multisets within element-pair types while relaxing atom consistency. Another matches incident bond assignments and then atoms injectively; each event touches two atoms, so half the incident cost, rounded up, is a lower bound. The stronger bound is retained. These relaxations can rule out improvement but need not be attainable.

Optimize the selected fragment's binary symmetries. Search additional discovery correspondences only when its recorded growth connectivity differs from final internal connectivity on either endpoint and the current witness remains unsettled. New starting correspondences must satisfy the saved discovery domains, anchors and accepted edges; subsequent binary symmetry choices are not rejected by those discovery constraints. Preparation, correspondence improvement and symmetry optimization share a five-second per-fragment deadline. Attaining a valid bound stops work immediately. Otherwise retain an actual best witness and incomplete status. An orbit minimum does not certify a minimum over discarded starting orbits.

Branch assembly binds local response operations to their shared branch choices. Final partition pieces own disjoint internal contributions. Weight-preserving global placements move endpoints while retaining those local costs. Ordinary boundaries remain fixed under the declared model. The sweep correction follows jointly reachable endpoint pairs through the saved action order, preserving dependence on variable internal events. When internal costs are constant, this is an endpoint-pair calculation, not a complete symmetry-assignment enumeration.

Window selection tracks a lower bound for each branch and an attained count U from a valid joint choice. If the smallest pre-sweep total L is attained by compatible fragment minima, its sweep residual gives $U \leq L+2$. Branches are examined only while their lower bounds can improve U; afterward, those with lower bounds

above $U + n$ are excluded. If local searches are unfinished or their minimizing choices conflict, the decoder retains their lower bounds and uses an actual witness count as U. The saved result records the retained branches, shared responses, window and whether the model minimum was proved.

For a requested event interval, the saved response searches its retained choices and returns a concrete mapping with its ordered choices, absence after complete exclusion, or unknown under a limit. Event counts and mappings can be replayed independently. Preparing a queryable response does not assert that every count in its numerical interval is attainable. Physical whole-reaction event counts can differ from the independent-boundary model and are recorded separately during external audits.

A.7 Indexed fragment reuse and execution

The reaction-local pool stores one original fragment per atom-set pair. A restricted saved witness proves a growth proposal when it covers the requested base atoms, uses available target atoms, respects locks and supports every required edge at the current tolerance. Atom-set containment alone is insufficient when occupancy or edge requirements have changed. An inverted index intersects atom, edge-support and fixed-assignment sets to find a valid proof. A miss invokes the ordinary existence query; no negative cache is used.

Growth continues with its own random proposals. Cut/trial streams are distinct and reproducible, and completion chooses its representatives independently of the temporary existence witness. The pool therefore does not replay a previous trajectory. Its accounted retention limit bounds cached records and index storage, not total stage memory. Returned search graphs expose their unique-fragment inventories directly to reduction.

Persistent execution shares compatible reaction snapshots at task boundaries. A permanent first snapshot and replaceable latest snapshot support concurrent readers; a validated first snapshot can be used when the latest stored version cannot be loaded. This is recovery of reusable data, not an alternative scientific algorithm. A single publication owner atomically replaces complete versions, and stale publishers cannot overwrite newer ones. Stored snapshot count is bounded per reaction rather than per cut or seed.

Separate stage APIs return typed matching, catalogue and decoding records. In-memory and aggregated file-backed transport preserve shared immutable arrays and fragment relations. Worker pools schedule those operations without changing their scope. For the reported batch protocol, a global matching/reduction barrier precedes decoding. Recovery retains completed outputs and consumed budgets; resource errors remain distinguishable from mapping absence. The main manuscript reports successful compute costs separately from failed attempts and external evaluation.

B Reproducibility and measurement scope

The repository's `manuscript/REPRODUCIBILITY.md` links source pins, immutable input identities, benchmark records and the evidence used to generate tables and figures. The public Python API documents direct matching, reduction, decoding and conditional queries. Figure building runs no searches. Publication records omit execution-site identifiers.

The paired Golden comparison uses SLAPMapper revision `ea248fd` and an Intel Xeon Gold 6442Y CPU. Exact source identities and verification-audit hashes are recorded in `evidence/paired_slap.json`. The paired comparison and full seed studies retain their respective MAPPA outputs and coverage counts as separate measurements.

The paired study records 1,839 MAPPA recoveries (99.352%), 1,794 expanded-SLAP recoveries (96.921%) and 1,620 standard bidirectional SLAP recoveries (87.520%). Expanded SLAP has 52 confirmed misses and five inconclusive evaluations; standard bidirectional SLAP has 230 misses and one inconclusive evaluation. Inconclusive results do not count as recoveries. Independent graph-matching evaluation agrees with all 3,683 completed expanded-SLAP direction verdicts. Default forward-only SLAP recovers 1,591 references (85.95%); the original paper reports 86.9% on its 1,758-reaction Golden version (9).

Paired CPU costs sum both directions and exclude loading, publication and external verification. The three-protocol comparison uses their 1,834 common completed reactions. A broader standard-SLAP/MAPPA

comparison contains 1,849 completed reactions, with mean search costs of 0.487 and 1.259 CPU seconds, respectively. It retains a slower standard-SLAP tail excluded from the three-protocol cohort. These timing subsets do not change the 1,851-reaction coverage denominator.

At the detected minimum, exact choice counts are available for 1,829 Golden reactions. The remaining 22 counts contribute bounds to full-dataset statistics. Minimum-set reference membership is established for 1,691 reactions and excluded for 158; two bounded queries account for the reported coverage range. These are separate from the completed membership evaluation of the wider minimum-plus-five window. The expected uniform-choice accuracy uses $1/K$ only when the reference is present, and bounds incomplete counts rather than assigning them a guessed value.

The main-text Golden measurements combine the saved one- and ten-seed matching passes with corrected completed reduction and decoding passes. They are not new uninterrupted end-to-end measurements. The matching passes predate indexed containment; the paper treats their seed scaling as an empirical workload comparison, not a cache ablation. EnzymeMap combines original successful tasks and successful recoveries with unchanged scientific settings. Performance corrections preserved the mapping relation and event model. CPU totals include computation, stage loading and record construction; publication, external audits and sampled earlier failures are separate. CPU models were not retained in the exported full-campaign summaries, so no hardware-normalized ranking follows.

Every accepted decoded-reference witness is checked for replay, injectivity, annotated assignments and an independently counted model total inside the saved window. Its physical total is stored separately. The Golden decoded audit uses exact indices; its catalogue audit permits endpoint chemical equivalence. EnzymeMap uses exact indices for both. A conclusive absence from the unchanged containing catalogue implies absence from the decoded subset; that inference is recorded separately from a direct negative decoded query. Bounded unknown queries are never relabeled as direct negatives.

The independent-fragment model and directly counted whole-reaction totals differ for 2,801 positive EnzymeMap directional witnesses and 32 ten-seed Golden witnesses. Both totals are retained in the audit records; decoded membership uses the saved model window.

Figure 7 uses the saved verified mapping counts and original detected minima in `evidence/reference_offsets.json.gz`. Each of the 49,825 reactions has a concrete offset. For recovered references, select the lowest model event count among the final saved forward/reverse verification witnesses; forward wins ties. The other 196 references use the saved anchored completions in `evidence/anchored_offsets.json`, with the same selection rule. No additional symmetry, starting-correspondence or branch-minimum search is performed. All counts include hydrogens under the same model; a selected hydrogen completion need not be optimal. The nine negative EnzymeMap values are anchored completions two events below the original detected minimum. These completions do not change independent coverage. Certificate and input identities retain the connection to the original audits.

The supplementary September studies below retain their original settings, evaluators and comparison cohorts. Their archived paired search timings use an Apple M4 Pro CPU; the cap comparison uses an AMD EPYC 9J14. Search, publication and event-query boundaries differ across those studies and are stated with each table. They are not pooled with the current main-text timings. The WBO-generation method for the archived coordinate inputs is not recorded in the available metadata, so regeneration of those matrices from molecular structures is not established.

C Archived September evaluations

The following studies used their recorded September implementations. They retain the original numerical results and comparison scopes; they are not reruns of the current algorithm or direct performance comparisons with Section 3.1.

Table A1 Archived evaluation configurations. Both use $b = 0.2$, $\tau_{iso} = 1$, explicit H, and uncut plus single-edge searches. One seed per cut and the sweep define the default setting. Archived Golden measurements cover sweep search; no-sweep controls retain only the uncut graph. A seed setting counts base-atom orderings per cut. Stored orientation uses the first saved endpoint as the base and the second as the target.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Bidirectional	Stored orientation
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

C.1 Archived datasets and comparator protocol

Golden inputs use formal bond orders and explicit H, with reference labels removed before search. Recovery requires the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count, and stereochemistry. These chemical colors define reference equivalence; they do not imply that element/WBO matching itself enforces every chemical attribute. Failures and unresolved cases remain in the denominator. This measures reference inclusion among returned alternatives, not their ranking.

LocalMapper provides the prior accuracy-SOTA baseline (1). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration. SLAP reported 86.9% on that dataset version, counting reference inclusion among its candidates (9). These literature scores are separate from our strict re-evaluation. We additionally apply the cut sweep introduced here to SLAP; "SLAP sweep" denotes this extension, not a published SLAP score. It is our strongest evaluated prior-method baseline for alternative recovery. Released RXNMapper, Chython, Indigo, and RDT provide further comparisons (14; 2; 5; 13).

The coordinate collection contains 18–137 explicit atoms per endpoint, including organometallic and multi-component structures. It was inspected during development and has no verified atom mappings: it is not a mapping-accuracy benchmark. The archived study compared the lowest event count found and the patterns achieving that count. MAPPA used supplied WBO matrices; SLAP used XYZ-derived binary graphs. Returned SLAP heavy-atom assignments received hydrogen refinement and were scored on the same WBO matrices as MAPPA. The comparison covers those saved witnesses, not all possible SLAP hydrogen assignments or a fresh current-implementation evaluation.

Released comparators received the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid endpoint chemistry, duplicate atom labels and other failures remained in the denominator. Binary SLAP was retained as the stronger default configuration; bidirectional comparisons used both binary and weighted modes. SLAP returns multiple candidates, whereas the other comparators return at most one explicit bijection. These results describe the evaluated interfaces and scoring rule, including its difference from RDT's chemical-change criterion. The repository records released models, versions, adapter settings and failure frequencies.

C.2 Archived evaluation configurations

Table A1 lists the archived settings. Search did not use reference or comparator correspondences. Full configurations, implementation versions, and reproduction scripts are provided in the repository (Appendix B).

C.3 Archived Golden reference recovery and search cost

In the archived campaign, the one-seed MAPPA configuration recovered 99.08% of Golden references, compared with 97.03% for SLAP sweep (Table A2 and figure A1). These scores belong to that campaign, separate from Section 3.1. The paired comparison includes 43 references recovered only by MAPPA and 5 only by

SLAP. LocalMapper is the strongest evaluated method returning one bijection. Additional MAPPA seeds yielded modest recovery gains at increasing search cost; two seeds did not improve recovery over one.

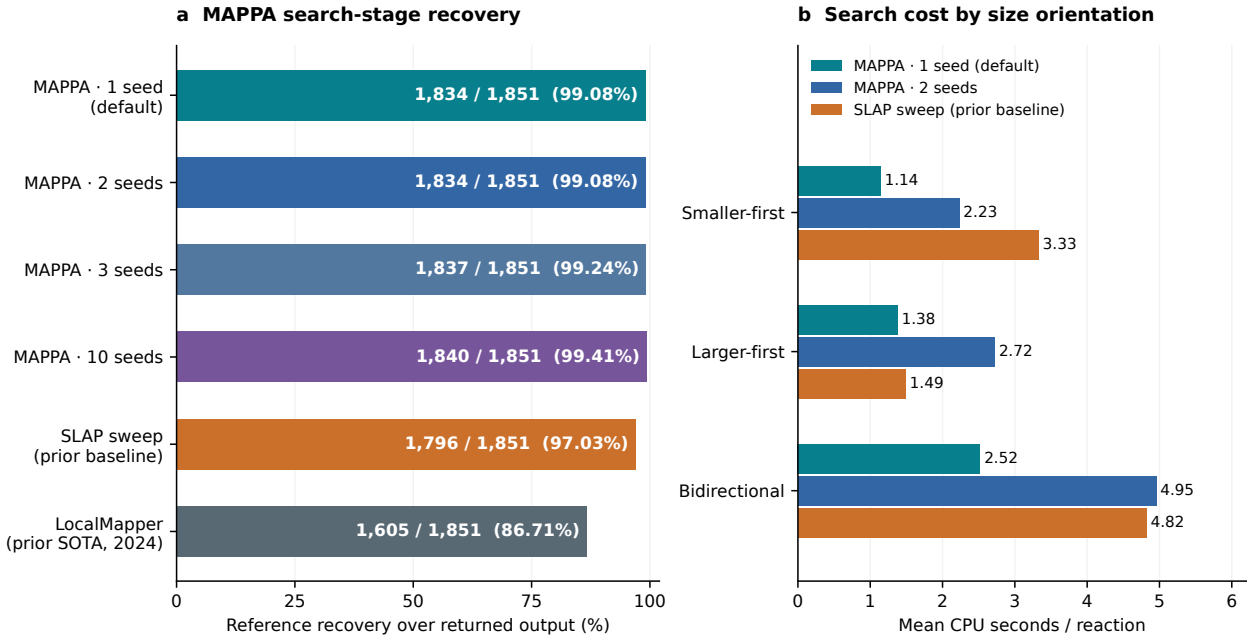


Figure A1 Archived Golden reference recovery and search cost. (a) Reference recovery over all 1,851 reactions. MAPPA and SLAP use our cut sweep; LocalMapper is the prior accuracy-SOTA baseline and returns one bijection. All scores use the archived evaluator. (b) Mean search CPU over the same 1,821 completed reactions, by orientation. Bond-event decoding is excluded. Full comparisons and timing scopes are in Tables A2 and A3.

The paired archived controls attribute 345 additional reference recoveries to the MAPPA sweep and 135 to the SLAP sweep. Without the sweep, one-seed MAPPA had lower coverage than bidirectional SLAP. This supports varying the growth constraints to expose alternatives excluded by greedy fragment extension (Section 2.3); symmetry expansion alone cannot restore discarded placements. These differences are not measurements of a current no-sweep control.

Smaller-first grows fragments from the reaction side with fewer explicit atoms (including H) and matches into the other side; larger-first reverses these roles. In the archived comparison, combining both directions added 31 recoveries over smaller-first alone, demonstrating complementary alternatives (Table A3).

In these archived measurements, one-seed MAPPA had a lower mean and median search cost than SLAP sweep, but a higher 95th percentile. Two seeds approximately doubled the mean cost without improving recovery. These Golden timings cover search, excluding reference verification and separate bond-event decoding. The two timing tables use different experiments and should be compared within their stated CPU and reaction sets; neither establishes a current-implementation speed ranking.

In the archived cap study, increasing the branch cap from 100 to 2,000 yielded no additional swept recoveries among cases resolved at both caps, while increasing cost and unresolved outcomes at higher seed counts (Table A8). The uncut search gained 19 confirmed recoveries. Thus, in that study a larger branch budget did not substitute for changing the growth conditions.

Table A2 Golden reference recovery and runtime in the archived evaluation. Recovery uses all 1,851 primary-evaluation records. MAPPA retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. MAPPA uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
MAPPA	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
MAPPA	2 seeds, sweep	Families	1,834 (99.08%)	1.619
MAPPA	3 seeds, sweep	Families	1,837 (99.24%)	2.265
MAPPA	10 seeds, sweep	Families	1,840 (99.41%)	6.993
MAPPA	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075[a]
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056[a]
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294[a]
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612[a]
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184[a]
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021[a]

Table A3 Archived recovery and runtime by search orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; CPU statistics are seconds per reaction over the same 1,821 completed reactions (Apple M4 Pro CPU). Smaller-first grows from the side with fewer explicit atoms, including H; larger-first exchanges the base and target. Equal-size ties use stored order and its reverse. Bidirectional output combines both searches, and its CPU cost is their sum. SLAP combines binary and weighted modes. CPU/SLAP compares means within each orientation. Timing scopes are specified in Appendix B.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
MAPPA, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
MAPPA, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

C.4 Archived bond-event comparison on 140 coordinate reactions

The archived coordinate study used one-seed MAPPA sweep search with a branch cap of 2,000 and returned complete mappings for all 140 reactions. At cap 100, cases 123 and 125 returned no complete mapping. The 2,000-branch configuration therefore supplies this comparison, while 100 remains the library default. These completion counts and the event-window results below have not been remeasured with the current implementation.

The archived MAPPA output matched or lowered the minimum event count found by both native SLAP and SLAP sweep in every reaction (Table A4 and figure A2). This does not imply recovery of every alternative: at equal minima, six SLAP-sweep patterns remained absent across four reactions. An equal count can describe different changed bonds, and a lower count alone does not establish a more plausible transformation.

Table A4 Archived minimum bond-event comparison on 140 reactions. MAPPA uses one-seed sweep search at cap 2,000. The first three columns count reactions by MAPPA's minimum event count relative to the comparator. The final column gives comparator minimum-event patterns recovered by MAPPA among reactions with equal minima. Counts refer to returned candidates, not proven global optima.

Compared with	MAPPA fewer	Equal	MAPPA more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

To compare alternatives above MAPPA's own minima, the archived study queried through one event above the higher of the two SLAP minima for each reaction. MAPPA retained 162 of 168 SLAP-sweep minimum-event patterns and 151 of 160 native-SLAP patterns within these windows. The output contains 300 distinct patterns, including 166 at MAPPA's own minima. Every saved family was exhausted or certified within its window; this establishes completeness for those retained families, not exhaustive structural search.

Table A5 Archived stage CPU time per reaction on the 140 coordinate reactions. Measurements use an Apple M4 Pro CPU. MAPPA uses one-seed sweep search at cap 2,000. These stage costs exclude MAPPA decoding and do not establish an end-to-end speed ranking. SLAP refinement covers its returned witnesses. Timing boundaries are specified in Appendix B.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The stage timings in Table A5 cover different outputs: MAPPA search retains families for subsequent decoding, while SLAP refinement operates on returned witnesses. They therefore do not establish an end-to-end speed ranking.

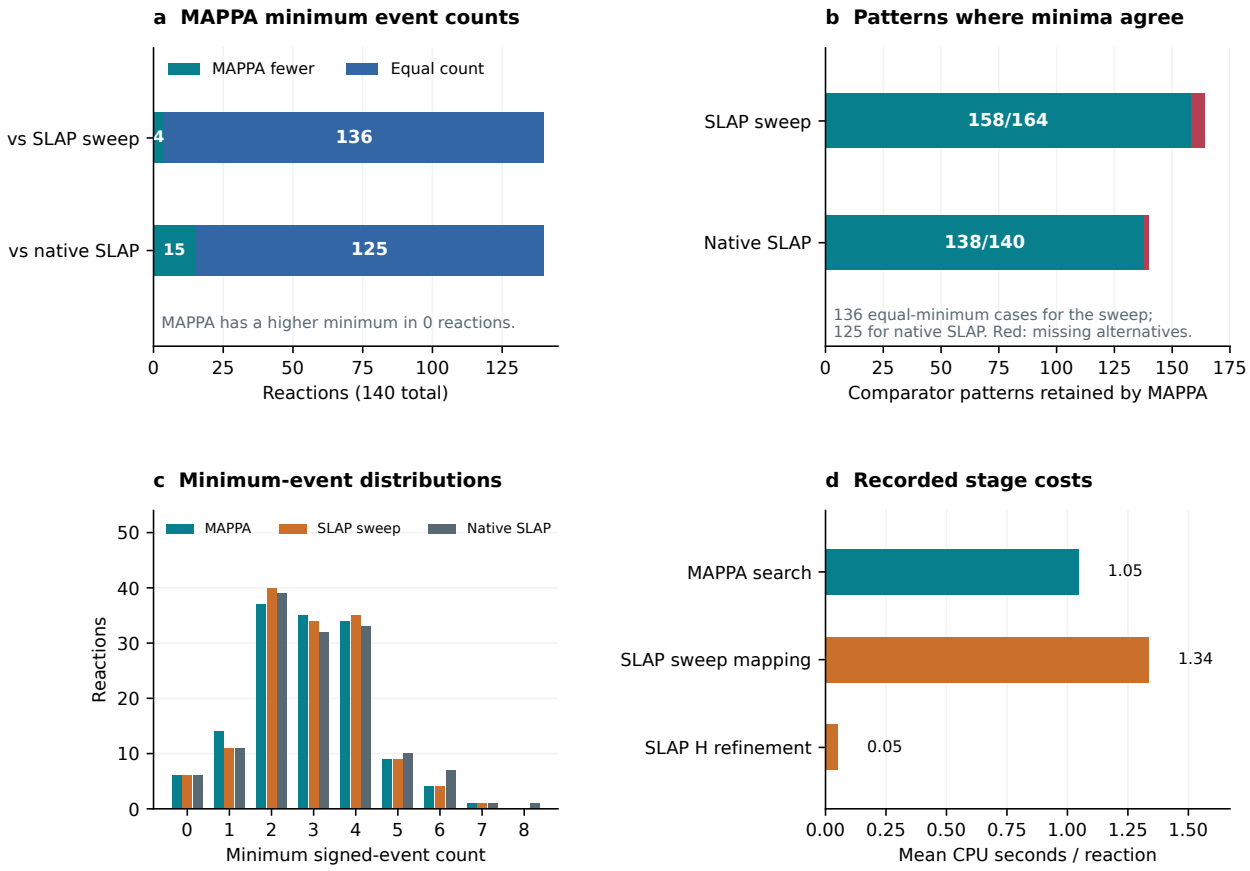


Figure A2 Archived bond-event comparison on 140 coordinate reactions. MAPPA uses one seed, the sweep, and cap 2,000. (a) Reactions with fewer or equal MAPPA events than each SLAP baseline. (b) Comparator minimum-event patterns recovered by MAPPA at equal minima. (c) Minimum-event count distributions. (d) Mean CPU for search and SLAP refinement, excluding MAPPA decoding (Table A5). Atom mappings are unverified; this compares bond events, not mapping accuracy.

C.5 Archived event-window query scope

For each coordinate reaction, the archived event-window limit was one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families were exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed atom-pair changes, modulo base permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness. This explicit whole-reaction query is separate from ordinary decoding into shared local fragment responses.

Table A6 Archived complete event-window queries on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

The query used event invariance to certify constant families and encoded remaining choices in Z3 (4). A family was certified by exhaustion, a valid lower bound excluding the window, or unsatisfiability after excluding all discovered classes. An exhausted resource budget remained incomplete. The explicit query adapter can simplify adjacent nested action groups by exact generator-membership checks: if $H \leq G$, then $HG = GH =$

G. This adapter simplification is separate from ordinary catalogue reduction in Appendix A.5. Completeness within an event window concerns the saved families, not exhaustive structural search.

C.6 Archived Golden nonrecoveries

Table A7 The eleven archived ten-seed Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms. This is not the nonrecovery set of the current one-seed validation.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The archived uncapped traces below demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

Two archived traces show how greedy extension excluded a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Growth preserved bonds that these correspondences require separating. Uncapped trajectories still missed both references, while reference-constrained fragment choices established their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposed the reference to matching; the single-edge sweep did not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries or traces of the current matcher. For the other five correspondence misses, the audit did not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. Independent evaluation on annotated coordinate reactions is needed to assess generalization beyond the development collection.

C.7 Archived branch-cap and CPU-time ablation on Golden

The archived study ran the twelve configurations in Table A8 on all 1,851 Golden reactions using an AMD EPYC 9J14 CPU. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the archived primary evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the archived

Table A8 Archived Golden branch-cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default setting.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
MAPPA uncut	100	1	1,489 (80.44)	0	0.040	0.085
MAPPA sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
MAPPA sweep	100	2	1,834 (99.08)	0	1.619	7.260
MAPPA sweep	100	3	1,837 (99.24)	0	2.265	10.556
MAPPA sweep	100	10	1,840 (99.41)	0	6.993	33.211
MAPPA uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
MAPPA sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
MAPPA sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
MAPPA sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
MAPPA sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

primary evaluation: two previously recovered cases are unresolved under this study's limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

C.8 Archived end-to-end timing on coordinate reactions

The archived September implementation completed search and event-window decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A9).

The historical study measured one-seed, cap-2,000 cut-sweep search followed by its event-window decoder on the Apple M4 Pro CPU. CPU times are reported per reaction. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, and candidate serialization; initial imports and reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Detailed timing records and reproduction instructions are provided with the code.

Table A9 Archived MAPPA search plus event-window queries. CPU seconds per completed reaction in the 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470